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PROSTATE CANCER EPIDEMIOLOGY

Uncovering Disparities in Diagnosis, Survival & Access Across the U.S.

SEER 22 / National Cancer Institute • 1999–2022 • 268,000+ Patient Records

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Introduction

- **Research Question**

  How have prostate cancer incidence, mortality, and survival rates varied across race, geography, and stage of diagnosis in the United States from 1999 to 2022?

- **Why It Matters**

  Prostate cancer is the most common non-skin cancer in American men — 1 in 8 will be diagnosed in their lifetime.

  Survival outcomes differ dramatically by race and geography.

  These differences are large, measurable, and preventable.

- **Dataset: SEER 22**

  Surveillance, Epidemiology & End Results Program

  268,000+ patient records • 51 states • 1999–2022

- **Key Variables**

  Race/ethnicity, year, cancer stage at diagnosis, incidence & mortality rates (per 100K), 5-year survival, and U.S. state of diagnosis

- **Notable Feature**

  Spans the 2012 USPSTF Grade D recommendation against routine PSA screening, enabling before/after analysis of how policy changes shaped national diagnosis patterns.

Methods

- **Data Cleaning & Preparation**

  Removed incomplete records and standardized race/ethnicity categories to SEER conventions

  Validated all rates against published benchmarks

- **Statistical Modeling**

  OLS regression stratified by race group

  Avoids false numeric ordering of nominal race variable

  Year-over-year trends modeled 1999–2022 per group

- **Survival Analysis**

  5-year relative survival stratified by stage

  Localized, Regional, Distant, and Unknown groups

- **Tools & Deliverable**

  Python • pandas • statsmodels • Plotly • Flask

  SQLite database • Interactive HTML Dashboard

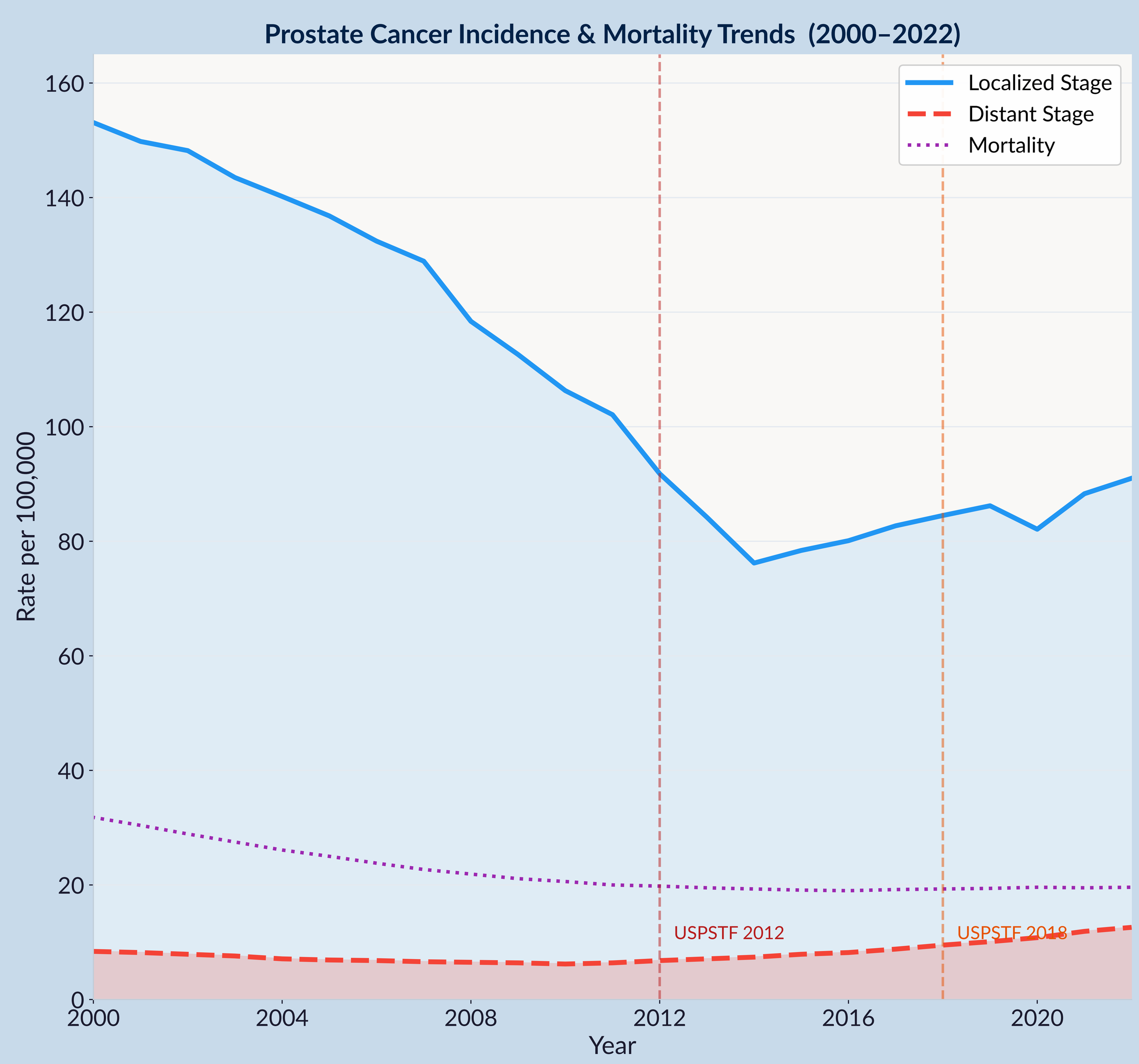
  Live filtering: race, state, year range, cancer stage

Finding 1 — 48% Incidence Decline, But Late-Stage Diagnoses Rising

153 → 91
per 100,000
Localized Incidence: 2000 → 2022

+50%
distant-stage
Rise in Late-Stage Diagnoses

- Post-USPSTF 2012: routine PSA screening discouraged nationally
- Fewer men screened → later-stage, harder-to-treat presentations

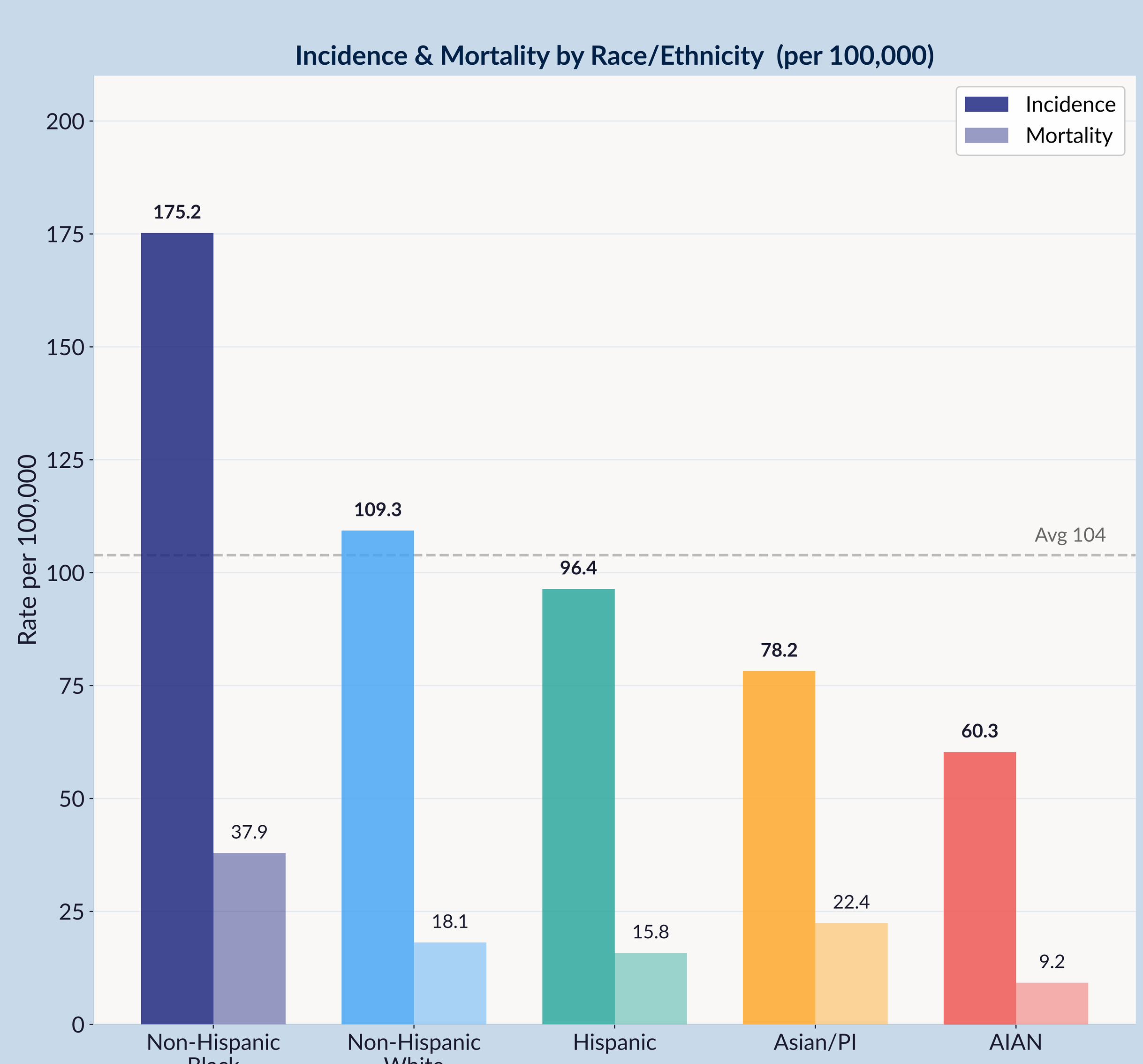


Finding 2 — Black Men: 60% Higher Incidence, 109% Higher Mortality

175.2
per 100,000
Black Men Incidence Rate

+109%
mortality gap
37.9 vs 18.1 per 100K

- Black men: 175.2/100K — 60% higher incidence than White men (109.3/100K)
- Disparity persists after controlling for age and stage at diagnosis

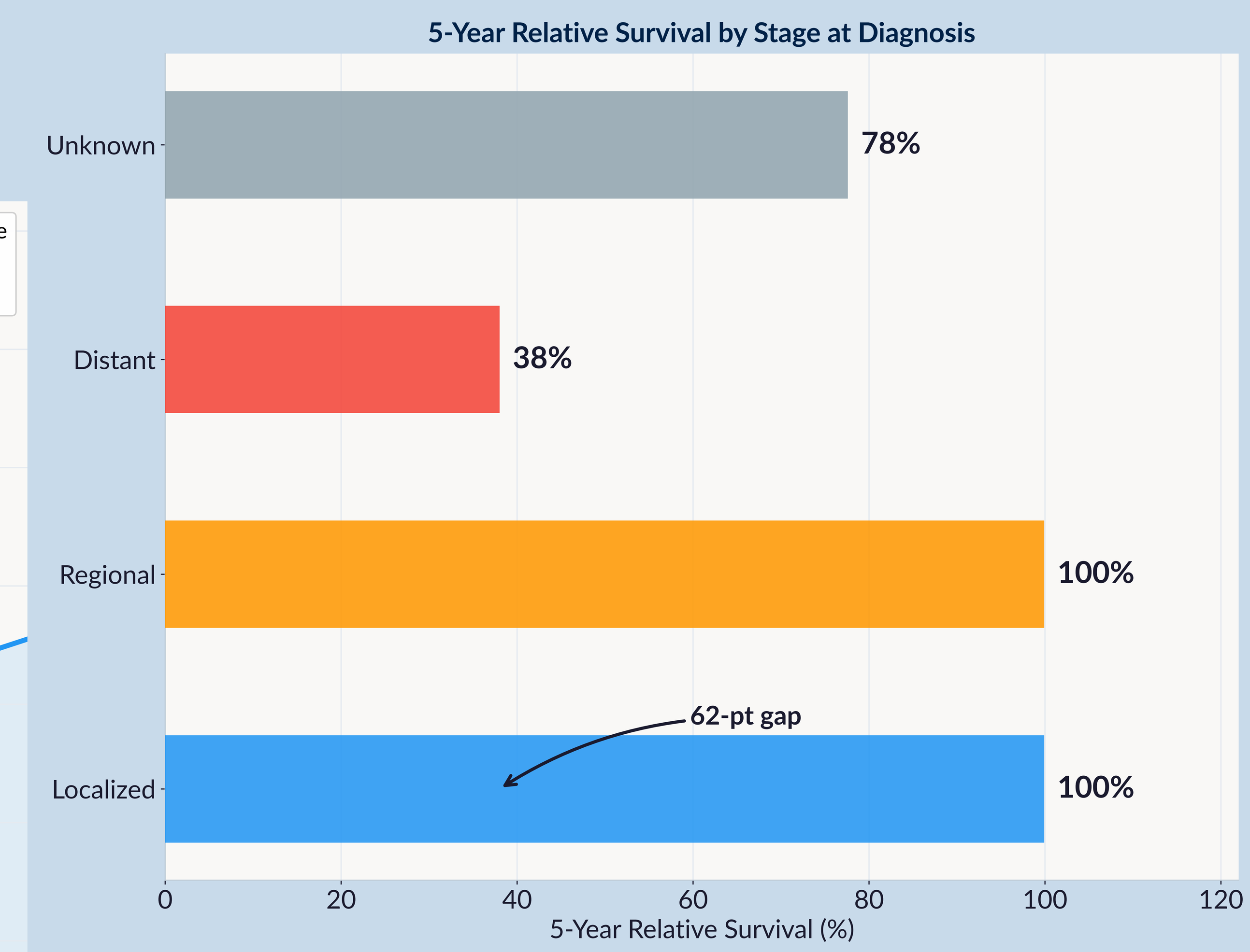


Finding 3 — Stage at Diagnosis Determines Whether You Survive

~100%
5-year survival
Localized Stage

38%
5-year survival
Distant Stage

- A 62-point survival gap between early and late-stage detection
- Rural & lower-income men significantly more likely to be diagnosed late



Conclusion

- **Key Findings**

  Incidence is falling nationally, but late-stage diagnoses are rising — masking a growing crisis within the headline. Early detection is being silently lost nationwide.

  Non-Hispanic Black men face 60% higher incidence and 109% higher mortality than White men — the starkest disparity in the data, persistent across age and stage.

  A 62-point survival gap separates localized from distant stage: where you live and who you are determines when you are diagnosed — and whether you survive.

- **Why It Matters**

  These disparities are large, measurable, and preventable with equitable access to early screening programs.

- **Limitations**

  SEER does not capture insurance status, income, or distance to care — key confounders for access-related disparities.

- **Future Directions**

  Integrate socioeconomic variables, rural access metrics, and insurance coverage data to build a full equity model.

Key Sources & Acknowledgements

Data: SEER 22, CDC USCS, PMC10720073, Kratzer et al. 2025 • avamcclinton@gmail.com